

Chapter 4A

Renal/Fluid Overview: Volume, Osmolality, Electrolytes, and Acid-Base Framework

Medvale Internal Medicine - Part V: Nephrology, Fluids, and Acid-Base

This chapter is an original Medvale study resource designed for clinical reasoning and future graph-RAG extraction. The uploaded renal volume/electrolyte chapter and Anki CSV were used as coverage signals only; wording, tables, algorithms, and diagrams are original.

Core promise of this chapter: When a patient has edema, oliguria, abnormal sodium, abnormal potassium, abnormal calcium, or a confusing blood gas, first decide whether the problem is one of effective arterial volume, water tonicity, kidney filtration, tubular response, or acid-base compensation. Most renal questions become tractable once those axes are separated.

1. Why this matters clinically

Fluid and electrolyte problems are common because every major organ system uses the kidney as a final common pathway. Heart failure, cirrhosis, nephrotic syndrome, sepsis, vomiting, diarrhea, adrenal disease, drugs, hyperglycemia, and intrinsic kidney disease can all present as a sodium disorder, a potassium emergency, edema, or acute kidney injury. The same serum sodium value can mean opposite things depending on volume status: a sodium of 124 mEq/L in a patient with diarrhea usually means salt and water depletion with more sodium than water lost, whereas the same sodium in heart failure usually means neurohormonal water retention from low effective arterial volume despite total-body fluid excess.

The safest first move is to separate **volume** from **tonicity**. Volume describes the amount of effective circulating extracellular fluid available to perfuse organs. Tonicity describes the osmotic force that shifts water into or out of cells, especially brain cells. Sodium concentration is mostly a marker of water balance, not a direct measurement of total-body sodium. Total-body sodium determines extracellular volume; water excess or deficit determines serum sodium concentration.

The kidney is both a sensor and an effector. A low-flow kidney avidly retains sodium and water through sympathetic tone, renin-angiotensin-aldosterone signaling, and antidiuretic hormone. A damaged tubule may fail to concentrate urine, waste sodium, generate casts, or excrete acid. Therefore, urine studies are not random add-ons; they are the kidney's report card on whether the physiologic response is appropriate.

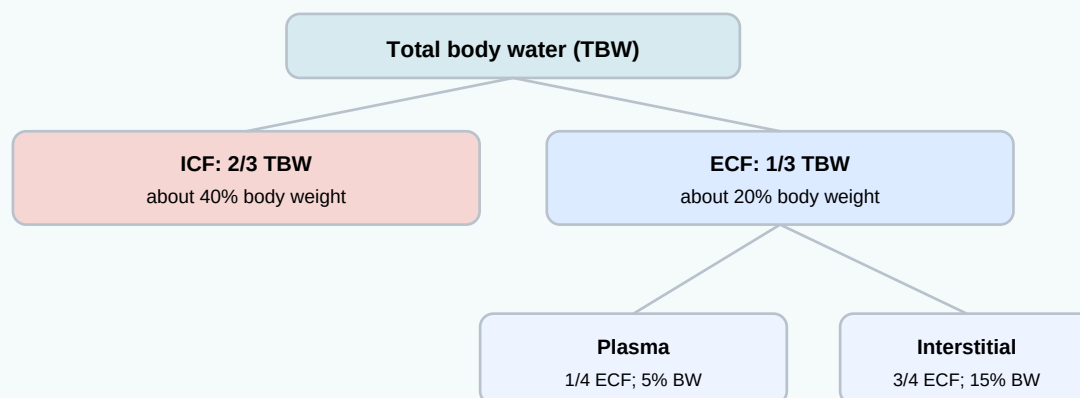
2. Core illness scripts

Script	Clinical picture	Renal/urine pattern	First danger
True volume depletion	Orthostasis, tachycardia, dry mucosa, low JVP, weight loss, possible cool extremities; causes include hemorrhage, vomiting, diarrhea, burns, diuretics, poor intake.	Low urine Na ⁺ often <20 mEq/L if tubules intact; high urine osmolality; BUN/Cr may rise disproportionately.	Shock, AKI, lactic acidosis.
Low effective arterial volume with edema	Edema or ascites but underfilled arterial circulation: heart failure, cirrhosis, nephrotic syndrome.	Neurohormonal sodium and water retention; urine Na ⁺ often low unless on diuretics or intrinsic renal disease.	Pulmonary edema, dilutional hyponatremia, kidney hypoperfusion.
Euvolemic hypotonic hyponatremia	No obvious edema or orthostasis; CNS symptoms if severe or acute; SIADH, adrenal insufficiency, hypothyroidism, drugs, low-solute intake.	Urine osmolality inappropriately >100 mOsm/kg if ADH active; urine Na ⁺ often >30 mEq/L in SIADH-like states.	Seizures from cerebral edema; osmotic demyelination if overcorrected.
Water deficit / hypernatremia	Thirst, delirium, weakness, dry mucosa; often impaired access to water or excess losses.	Urine should concentrate >600-800 mOsm/kg unless diabetes insipidus, osmotic diuresis, or renal concentrating defect.	Brain shrinkage acutely; cerebral edema if corrected too fast.
Acid-base overlay	Dyspnea, altered mental status, shock, renal failure, intoxication, vomiting/diarrhea, COPD, sepsis, DKA.	Kidney should alter ammonium, bicarbonate handling, chloride, and potassium over hours to days.	Missing a mixed disorder or toxic ingestion.

3. Body water and compartment logic

A practical adult estimate is the 60-40-20 rule: total body water is about 60% of body weight in many adult men, about 50% in many adult women, and lower with older age or higher adiposity. Intracellular fluid is roughly two-thirds of total body water, or about 40% of body weight. Extracellular fluid is roughly one-third of total body water, or about 20% of body weight. Within extracellular fluid, about one-quarter is plasma and three-quarters is interstitial fluid. These numbers are approximations, but they explain why a liter of isotonic saline does not stay entirely intravascular and why free water distributes far beyond the plasma space.

Original compartment map: where administered water and sodium distribute



Pearl: sodium content largely controls ECF volume; water movement follows tonicity across cell membranes.

The clinical distribution of a fluid depends on its effective osmoles. Isotonic saline primarily expands the extracellular compartment because sodium and chloride remain mostly extracellular. Dextrose in water becomes electrolyte-free water after glucose metabolism and distributes across total body water; only a small fraction remains in plasma. Balanced crystalloids, such as lactated Ringer solution or Plasma-Lyte, also expand extracellular fluid but provide less chloride load than normal saline. Colloid solutions generate oncotic force in theory, but benefit depends on capillary integrity and the clinical setting.

Numerical anchor: In a 70-kg adult with TBW near 42 L, ICF is roughly 28 L and ECF is roughly 14 L. Plasma volume is roughly 3 to 3.5 L. A 1-L free water load has a much smaller immediate plasma effect than a 1-L isotonic crystalloid load, because free water distributes across TBW rather than staying within ECF.

4. Effective arterial blood volume

Effective arterial blood volume (EABV) is the perfusion signal sensed by baroreceptors and the kidney. It is not the same as total-body water and not the same as visible edema. A patient with severe heart failure may have liters of excess interstitial fluid but a low EABV because forward arterial flow is poor. A patient with cirrhosis may have ascites and edema but low EABV because splanchnic vasodilation and low oncotic pressure reduce effective perfusion. A patient with nephrotic syndrome may be edematous because hypoalbuminemia and renal sodium retention favor interstitial expansion while the arterial circulation is still perceived as underfilled.

The kidney's response to low EABV is conservation: increased sympathetic tone, renin release, angiotensin II-mediated efferent arteriolar tone and proximal sodium reabsorption, aldosterone-mediated distal sodium reabsorption and potassium secretion, and ADH-mediated water retention. This response is adaptive during hemorrhage but maladaptive in chronic heart failure or cirrhosis because it worsens congestion and dilutional hyponatremia.

Volume status is a perfusion question, not just an edema question

Volume category	Physical/lab clues	Common mechanisms	Therapeutic instinct
Hypovolemic	Orthostasis, tachycardia, flat neck veins, weight loss, low urine output; hemoconcentration may appear after equilibration.	Hemorrhage, GI loss, poor intake, osmotic diuresis, burns, third spacing, excessive diuretics.	Restore EABV with isotonic crystalloid or blood products when indicated; monitor urine output and perfusion.
Euvolemic	No major edema or orthostasis; neurologic symptoms may reflect tonicity rather than volume.	SIADH-like ADH excess, adrenal insufficiency, hypothyroidism, low solute intake, primary polydipsia.	Treat cause; control free water; avoid rapid sodium correction.

Volume category	Physical/lab clues	Common mechanisms	Therapeutic instinct
Hypervolemic	Edema, ascites, pulmonary crackles, elevated JVP, weight gain, sometimes low sodium from water retention.	Heart failure, cirrhosis, nephrotic syndrome, kidney failure, iatrogenic sodium/water excess.	Sodium restriction, loop diuretics, disease-specific therapy, dialysis/ultrafiltration if refractory or life-threatening.

5. Intake, output, oliguria, and daily weights

Volume status is difficult to determine from a single examination. Skin turgor and mucous membranes can mislead, edema may coexist with low EABV, and intake/output records can be incomplete. Daily weights are often the most reliable trend when measured consistently. A change of 1 kg approximates a 1-L change in water balance, although third-spacing and nutritional changes can obscure the relationship.

Urine output is one of the most important early safety endpoints. A commonly used adult target is at least 0.5 mL/kg/hour, which is about 35 mL/hour in a 70-kg adult. Oliguria generally means urine output below about 400-500 mL/day in adults, whereas anuria is often below about 100 mL/day. Low urine output can be pre-renal hypoperfusion, intrinsic kidney injury, or post-renal obstruction; do not assume dehydration until obstruction and congestion have been considered.

Maintenance needs vary with fever, respiratory rate, burns, drains, diarrhea, diuresis, and renal/cardiac function. A classic maintenance estimate for adults is roughly 25-30 mL/kg/day, adjusted downward in heart failure, advanced CKD, cirrhosis with hyponatremia, and oliguric AKI. Insensible losses are often near 500-1000 mL/day, rising with fever, tachypnea, sweating, open wounds, or high ambient temperature.

6. Osmolality, tonicity, and sodium concentration

Serum osmolality counts solute particles in plasma. Tonicity counts the effective osmoles that cannot freely cross cell membranes and therefore shift water between ECF and ICF. Urea contributes to measured osmolality but is relatively ineffective for tonicity because it crosses cell membranes. Sodium salts and glucose are clinically important effective osmoles. This is why severe hyperglycemia can lower measured serum sodium by pulling water from cells into the extracellular space while raising effective tonicity.

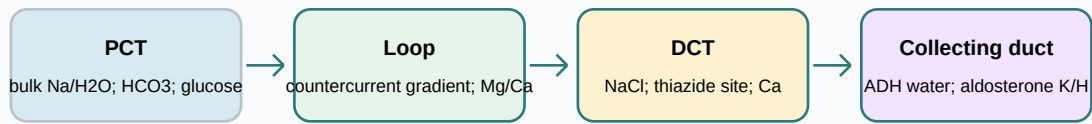
Useful formulas: calculated serum osmolality = $2 \times \text{Na}^+ + \text{glucose}/18 + \text{BUN}/2.8$, using mg/dL for glucose and BUN. Effective osmolality, or tonicity, is often approximated as $2 \times \text{Na}^+ + \text{glucose}/18$. Osmolar gap = measured serum osmolality - calculated serum osmolality. A high osmolar gap suggests unmeasured osmoles such as toxic alcohols, mannitol, propylene glycol, or other small solutes, but interpretation depends on timing and metabolism.

Osmolality tests answer specific questions

Question	Key measurement	Interpretation
Is the sodium disorder real?	Measured serum osmolality	Low osmolality supports true hypotonic hyponatremia. Normal or high osmolality suggests pseudohyponatremia or osmotic shift from glucose/mannitol-like solutes.
Is ADH active?	Urine osmolality	<100 mOsm/kg means maximally dilute urine and suppressed ADH. >100 means ADH effect or impaired dilution. >600-800 during hypernatremia means appropriate concentration.
Is the kidney retaining sodium?	Urine sodium, sometimes urine chloride	Low urine Na^+ suggests sodium avidity from low EABV if tubules are intact. High urine Na^+ suggests renal salt loss, diuretics, adrenal insufficiency, or SIADH-like physiology.
Is there an unmeasured osmole?	Osmolar gap	Gap >10-15 mOsm/kg is a clue, not a diagnosis; pair with anion gap, lactate, ketones, renal function, and exposure history.

7. Basic nephron physiology for clinical reasoning

Nephron decision axis: match the disorder to the segment



Common localization clues: FeNa/FeUrea and casts for AKI; urine osm/Na for water disorders; urine pH and K⁺ for acid-base/tubular disorders.

The proximal tubule reabsorbs most filtered sodium and water iso-osmotically, along with bicarbonate, glucose, amino acids, phosphate, and uric acid handling. Proximal dysfunction therefore produces broad losses: bicarbonaturia in proximal RTA, phosphaturia, glycosuria with normal serum glucose, aminoaciduria, and uricosuria in Fanconi-like syndromes.

The loop of Henle builds the medullary gradient. The thick ascending limb reabsorbs sodium, potassium, and chloride without water, diluting tubular fluid while maintaining the gradient needed for water reabsorption later. Loop diuretics interrupt this gradient, increasing salt and water excretion and also increasing calcium and magnesium losses. The distal convoluted tubule fine-tunes sodium chloride reabsorption and calcium handling; thiazides act here and tend to increase calcium reabsorption.

The collecting duct is the final regulated segment. ADH inserts aquaporin-2 water channels, allowing water reabsorption down the medullary gradient. Aldosterone increases epithelial sodium channel activity and Na/K-ATPase function, promoting sodium retention while secreting potassium and hydrogen ions. This is why hyperaldosteronism causes hypertension, hypokalemia, and metabolic alkalosis, whereas hypoaldosteronism or aldosterone resistance causes hyperkalemia and non-anion-gap metabolic acidosis.

8. Hormonal control loops

System	Trigger	Main action	Clinical signature
RAAS	Low renal perfusion pressure, sympathetic activation, low distal NaCl delivery.	Angiotensin II constricts efferent arteriole and increases proximal sodium reabsorption; aldosterone retains sodium distally and secretes K ⁺ /H ⁺ .	Low urine sodium in low EABV; secondary hyperaldosteronism in heart failure/cirrhosis; hypokalemic alkalosis if excessive.
ADH	High tonicity; also nausea, pain, stress, low EABV, drugs.	Aquaporin insertion in collecting duct increases free water reabsorption.	Concentrated urine in hyponatremia when ADH is inappropriately high; dilute urine in central DI or primary polydipsia.
Aldosterone	Angiotensin II, hyperkalemia, ACTH minor.	Sodium retention, potassium secretion, hydrogen secretion in collecting duct.	Hypokalemia/alkalosis when high; hyperkalemia/acidosis when low or resisted.
Natriuretic peptides	Atrial/ventricular stretch.	Increase natriuresis and oppose RAAS/SNS signaling.	BNP/NT-proBNP support congestion assessment but are not specific for isolated volume overload.
PTH, vitamin D, FGF23	Low calcium, high phosphate, CKD-mineral axis disruptions.	Regulate calcium/phosphate exchange across bone, kidney, and gut.	CKD: phosphate retention, low calcitriol, secondary hyperparathyroidism, renal osteodystrophy risk.

9. Initial approach to electrolyte abnormalities

Electrolyte interpretation begins with danger, not memorization. Ask whether the abnormality can kill the patient immediately: potassium abnormalities through arrhythmia, calcium abnormalities through seizures or arrhythmia, sodium abnormalities through brain swelling or osmotic injury, magnesium abnormalities through torsades or neuromuscular instability. Stabilize first, then calculate deficits and causes.

Electrolyte	Immediate danger pattern	Initial framework	High-yield anchors
Sodium	Seizure/coma if acute severe hyponatremia; delirium/weakness in hypernatremia.	Check serum osm -> volume status -> urine osm -> urine Na+.	Hyponatremia is usually water excess relative to sodium; hypernatremia is usually water deficit. Avoid overly rapid correction.
Potassium	ECG changes, ventricular arrhythmia, paralysis.	Confirm value, check ECG, acid-base status, kidney function, medications, aldosterone axis.	Insulin/beta-agonist shift K+ intracellularly; calcium stabilizes myocardium but does not lower K+; definitive removal requires kidney/GI/dialysis.
Calcium	Tetany/seizure/QT prolongation if low; stones/bones/psychiatric groans/short QT if high.	Correct for albumin or measure ionized Ca2+; check PTH, phosphate, magnesium, vitamin D, kidney function.	Low Mg can suppress PTH. Hypercalcemia often causes nephrogenic DI-like water loss and hypovolemia.
Phosphate	Weakness, rhabdomyolysis, respiratory failure if severe low; hypocalcemia and vascular calcification when high.	Think refeeding, DKA treatment, alcohol use, respiratory alkalosis, CKD, tumor lysis.	Phosphate binds calcium; kidney excretion is central.
Magnesium	Torsades, seizures, refractory hypoK/hypoCa if low; hypotension/bradycardia if high.	Check GI loss, diuretics, alcohol use, PPIs, aminoglycosides, amphotericin, cisplatin, kidney failure.	Correct magnesium when potassium or calcium will not correct.

10. Initial approach to acid-base disorders

Acid-base analysis should be mechanical. First determine pH. Then identify whether the primary process is metabolic or respiratory by the direction of bicarbonate and PaCO₂. Third, determine whether compensation is appropriate. Fourth, calculate the anion gap if metabolic acidosis is present. Fifth, compare the change in anion gap to the change in bicarbonate to detect mixed disorders. A single patient can have DKA, vomiting, renal failure, and sepsis at the same time; the blood gas should be read as an integrated physiologic story.

Tool	Formula / anchor	Use
Anion gap	$\text{Na}^+ - (\text{Cl}^- + \text{HCO}_3^-)$; often normal about 8-12 mEq/L depending on lab.	Classifies metabolic acidosis into high-gap vs non-gap patterns. Correcting for low albumin may reveal a hidden gap.
Winter formula	Expected $\text{PaCO}_2 = 1.5 \times \text{HCO}_3^- + 8 \pm 2$.	Tests respiratory compensation for metabolic acidosis. Higher PaCO ₂ implies concurrent respiratory acidosis; lower implies respiratory alkalosis.
Metabolic alkalosis	Think chloride-responsive vs chloride-resistant; urine chloride is often more useful than urine sodium.	Vomiting/NG suction and remote diuretics are often chloride-responsive; mineralocorticoid excess is chloride-resistant.
Respiratory disorders	Acute vs chronic compensation differs; kidneys need time to retain or excrete bicarbonate.	COPD/OHS may chronically elevate bicarbonate; sudden worsening adds acute respiratory acidosis.
Osmolar gap + anion gap	Gap pattern changes over time as toxic alcohols metabolize into acids.	Suspect methanol, ethylene glycol, propylene glycol, or other unmeasured osmoles when exposure and labs fit.

11. Diagnostic approach: a unified first-pass algorithm

First-pass renal/fluid algorithm

1. Is the patient unstable ?

Airway, breathing, circulation, ECG, glucose; give isotonic crystalloid if shock.

2. What is the effective arterial volume?

Orthostasis/JVP/edema/lungs + trend in weight, urine output, creatinine.

3. What is the tonicity problem?

Serum osm; compare sodium with glucose and other effective osmoles.

4. What does the kidney say?

Urine osm, urine sodium, urine chloride, casts, protein, blood.

5. Is there an acid-base overlay?

pH -> primary process -> compensation -> anion gap/delta gap.

In unstable patients, do not wait for a perfect diagnosis. Treat shock, severe hyperkalemia, seizure-level hyponatremia, hypoglycemia, hypoxemia, and airway failure immediately while collecting targeted data. In stable patients, the best first renal panel includes serum electrolytes with bicarbonate, BUN/creatinine, glucose, calcium/magnesium/phosphate when relevant, serum osmolality for sodium disorders, urinalysis with microscopy, urine sodium and osmolality, and urine chloride when alkalosis is present. A renal ultrasound enters early when obstruction is plausible, when AKI is unexplained, or when kidney size/chronicity matters.

12. Diagnostic traps

Trap	Why it misleads	Safer interpretation
Treating edema as hypervolemia in every sense	Edema can coexist with low effective arterial volume.	Ask whether the arterial circulation is underfilled: heart failure, cirrhosis, nephrotic syndrome.
Assuming hyponatremia means sodium depletion	Most hyponatremia reflects water excess relative to sodium.	Use serum osm, volume status, urine osm, and urine sodium before choosing saline vs water restriction.
Ignoring glucose when sodium is low	Hyperglycemia lowers measured sodium by osmotic water shift.	Correct/interpret sodium in the context of glucose and effective osmolality.
Overtrusting urine sodium on diuretics	Diuretics can raise urine sodium despite low EABV.	Use timing, urine chloride, FeUrea, clinical context, and medication history.
Rapid correction of chronic hyponatremia	Brain adaptation makes chronic hyponatremia vulnerable to osmotic demyelination.	Set explicit correction goals and check sodium frequently during active therapy.
Missing obstruction in oliguria	Post-renal AKI may have minimal early lab clues.	Bladder scan/Foley assessment and renal ultrasound when plausible.
Calling compensation a second disease too early	Expected compensation can look abnormal unless calculated.	Apply compensation formulas before diagnosing mixed disorders.

13. High-yield exam clues

- **Low urine sodium with low urine output** often means intact tubular sodium retention from low EABV, but diuretics, CKD, and adrenal disease can break the pattern.
- **Urine osmolality below 100 mOsm/kg** in hyponatremia points toward suppressed ADH, classically primary polydipsia or low solute intake; higher values mean ADH is active or dilution is impaired.
- **Hypernatremia with dilute urine** suggests diabetes insipidus or a concentrating defect; hypernatremia with concentrated urine suggests intact ADH response and inadequate water intake or extrarenal loss.

- **Hypokalemia with metabolic alkalosis** asks for blood pressure and urine chloride: vomiting/remote diuretics vs diuretics/mineralocorticoid excess.
- **Hyperkalemia with non-anion-gap metabolic acidosis** suggests type 4 RTA/hypoaldosteronism, kidney failure, or drugs impairing RAAS/aldosterone signaling.
- **Low magnesium** can make hypokalemia and hypocalcemia refractory until magnesium is repleted.
- **Albumin changes total calcium**; ionized calcium is the physiologically active value, especially in critical illness or acid-base shifts.
- **A normal pH does not exclude severe acid-base disease**; paired metabolic acidosis and respiratory alkalosis can normalize pH while the patient is very ill.

14. Original quick-reference tables

Fluid selection by physiologic purpose

Fluid	Main effective distribution	Typical use	Caution
0.9% saline	ECF expansion; chloride-rich.	Hypovolemic shock, initial resuscitation, hypochloremic metabolic alkalosis.	Large volumes can worsen hyperchloremic metabolic acidosis and congestion.
Balanced crystalloid	ECF expansion with lower chloride burden.	Many resuscitation settings, perioperative/ICU fluids.	Contains potassium in small amounts; use judgment in severe hyperkalemia or specific compatibility issues.
D5W	Electrolyte-free water after glucose metabolism; distributes across TBW.	Free water replacement in hyponatremia once hemodynamically safe.	Poor resuscitation fluid; can worsen hyponatremia/hyperglycemia.
0.45% saline	Hypotonic sodium-containing water; part ECF, part ICF after equilibration.	Selected hyponatremia/free water deficits when some sodium is desired.	Can worsen hyponatremia or cerebral edema in vulnerable patients.
3% saline	Hypertonic ECF expansion; raises tonicity.	Severe symptomatic hyponatremia.	Requires careful dosing and frequent sodium checks.

Urine clues as pathophysiology

Finding	Most useful next thought
BUN/Cr >20 with low urine Na ⁺	Pre-renal physiology is possible, but GI bleed, catabolic state, steroids, and low muscle mass can distort interpretation.
Muddy brown granular casts	Acute tubular injury/necrosis pattern; correlate with ischemia, sepsis, toxins, pigment injury.
RBC casts or dysmorphic RBCs	Glomerulonephritis until proven otherwise.
WBC casts	Pyelonephritis or interstitial nephritis, depending on symptoms and drug exposure.
Heavy proteinuria and edema	Nephrotic physiology: quantify proteinuria, albumin, lipids, renal function, and systemic causes.
Urine chloride low in alkalosis	Chloride-responsive alkalosis such as vomiting or remote diuretics.

15. Practice questions

Question 1. A 72-year-old with heart failure has Na⁺ 126 mEq/L, bilateral crackles, edema, high JVP, and low urine sodium. What single physiologic phrase best explains the sodium abnormality?

Answer. Low effective arterial blood volume causing neurohormonal ADH-mediated water retention despite total-body fluid excess. The kidney is sodium avid because it senses underfilling.

Question 2. A patient with acute gastroenteritis is dizzy when standing, tachycardic, and has creatinine rise with urine Na⁺ 8 mEq/L. What is the initial fluid logic?

Answer. Restore EABV with isotonic crystalloid while monitoring perfusion, urine output, and electrolytes. The low urine sodium is an appropriate sodium-retaining response if tubules are intact.

Question 3. A sodium of 122 mEq/L is found in a patient with glucose 820 mg/dL and measured serum osmolality is high. Why is this not ordinary hypotonic hyponatremia?

Answer. Glucose is an effective osmole that pulls water from cells into ECF, lowering measured sodium while increasing tonicity. Management centers on hyperglycemic crisis and careful fluid/electrolyte correction.

Question 4. A patient with Na⁺ 118 mEq/L has urine osmolality 60 mOsm/kg. What does this say about ADH?

Answer. ADH is appropriately suppressed or absent; think primary polydipsia or low-solute intake rather than SIADH, which typically has inappropriately concentrated urine.

Question 5. A patient with metabolic acidosis has HCO₃⁻ 12 mEq/L and PaCO₂ 38 mmHg. What is the key acid-base concern?

Answer. Winter formula predicts PaCO₂ about 26 +/- 2. A measured PaCO₂ of 38 is too high, indicating concurrent respiratory acidosis or ventilatory failure on top of metabolic acidosis.

16. Graph-RAG extraction map

The table below is intentionally structured for concept-node extraction. Each row can become a node or edge candidate with source-context links, aliases, numerical anchors, and relationship labels.

Concept node	Canonical description	Useful edges	Numerical anchors / attributes
Total body water	Water content distributed across intracellular and extracellular compartments.	depends_on -> age/sex/body composition; partitions_into -> ICF/ECF	~60% male body weight; ~50% female body weight; lower with age/adiposity.
Extracellular fluid	Sodium-rich compartment that includes plasma and interstitial fluid.	expanded_by -> isotonic sodium fluids; determines -> edema/plasma volume	~1/3 TBW; ~20% body weight; plasma ~1/4 ECF.
Effective arterial blood volume	Perfusion signal sensed by kidney/baroreceptors, not identical to total fluid volume.	low_EABV activates -> RAAS/ADH/SNS; low_EABV causes -> sodium avidity	Clinical rather than single numeric variable; inferred from perfusion, JVP, edema, urine Na.
ADH	Posterior pituitary hormone that increases collecting duct water permeability.	stimulated_by -> high tonicity/low EABV/nausea/pain; causes -> concentrated urine/water retention	Urine osm >100 mOsm/kg implies ADH effect in most hyponatremia contexts.
RAAS	Hormonal cascade increasing sodium retention and vascular tone during perceived underfilling.	activated_by -> low renal perfusion; increases -> aldosterone; promotes -> K/H secretion	Low urine Na ⁺ often <20 mEq/L when sodium avid and tubules intact.
Hyponatremia framework	Serum sodium below normal; usually water excess relative to sodium.	classified_by -> serum osm/volume/urine osm/urine sodium	Severe often <120; correction limits require frequent monitoring.
Hypertatremia framework	Serum sodium above normal; usually water deficit relative to sodium.	caused_by -> water loss/impaired access/DI/osmotic diuresis/sodium gain	Na ⁺ >145 mEq/L; urine should concentrate if ADH axis intact.
Anion gap	Screen for unmeasured anions in metabolic acidosis.	high_in -> lactate/ketoacids/renal failure/toxins; paired_with -> delta gap	AG = Na - (Cl + HCO ₃); normal often ~8-12 depending lab.
Urine osmolality	Measures kidney water concentration response.	low_in -> suppressed ADH; high_in -> ADH effect/concentrating response	<100 dilute; >600-800 appropriately concentrated during water deficit.
Urine sodium	Measures renal sodium handling in context.	low_in -> sodium avidity; high_in -> renal salt loss/SIADH/diuretics	<20 often sodium avid; >30-40 often renal loss or SIADH-like state.

17. References and accuracy notes

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- Uploaded Medvale coverage sources: renal volume/electrolyte/acid-base chapter PDF and Anki card CSV supplied by the user for topic coverage only.

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